

QXU4002 – Chemistry for Materials

Question Set 10

1) Have a look at the Lecture Notes Set 6 and ask questions of what you do not understand.

2) The reaction $A(g) \rightleftharpoons B(g)$ is catalysed by a solid, C (s). The pressure of A vs time is shown in the table below, with arbitrary units. The adsorption of A on C follows Langmuir behaviour.

PA (Pressure units)	Time (time units)
27.5	5
25.0	10
22.5	15
20	20
17.5	25
15	30

a) Write an equation for the rate of the reaction, showing the values of all constants in arbitrary units. You may want to use the aid of graphs in calculating your answer.

The reaction follows Langmuir behaviour, then:

$$rate = kf = \frac{kKP}{1 + KP}$$

$$-\frac{dP_A}{dt} = \frac{kKP_A}{1 + KP_A} \xrightarrow{\text{Integrating}} \frac{1}{K} \ln \frac{P_A^0}{P_A} + P_A^0 - P_A = kt$$

If adsorption of A is strong or P_A is high, then $1 \ll KP$

$$rate = kf = \frac{kKP}{1 + KP} \approx \frac{kKP}{KP} = k$$

$$-\frac{dP_A}{dt} = k \xrightarrow{\text{Integrating}} P_A^0 - P_A = kt \quad \longrightarrow \quad P_A = P_A^0 - kt$$

P_A vs t is a linear equation

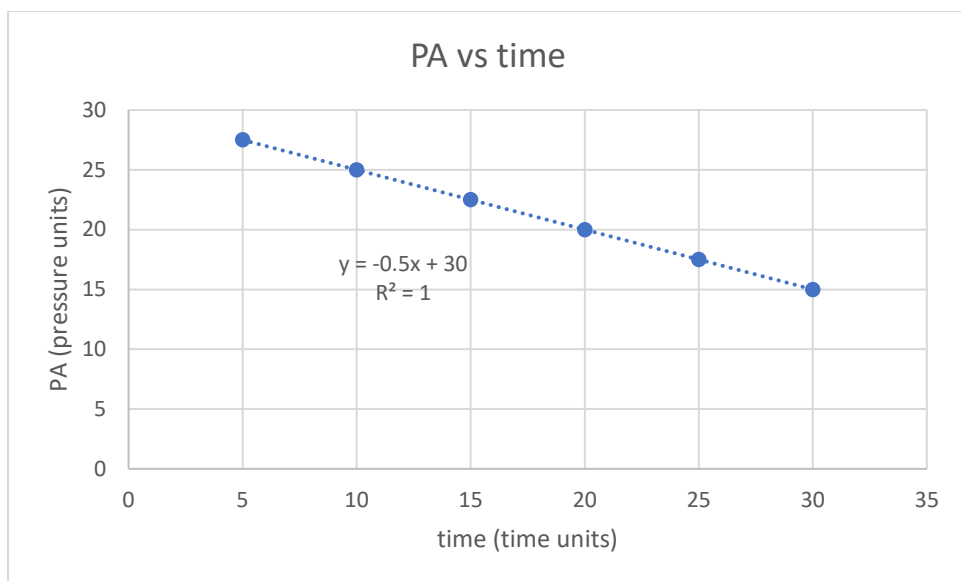
If adsorption of A is weak, or P_A is low: then, $1 \gg KP$

$$rate = kf = \frac{kKP}{1 + KP} \approx \frac{kKP}{1} = kKP$$

$$-\frac{dP_A}{dt} = kKP_A \xrightarrow{\text{Integrating}} \ln \frac{P_A^0}{P_A} = kKt \quad \longrightarrow \quad \ln P_A = \ln P_A^0 - kKt$$

$\ln P_A$ vs t is a linear equation

For the data provided in the table P_A vs t fits a linear equation



Where $P_A = 30 - 0.5t$, and $k = 0.5 \text{ (Pressure units)} \times \text{(time units)}^{-1}$

b) What can you tell about the interactions between A and C given the data above?

Since data of P_A vs t fit a linear equation, according to the Langmuir equation, that means adsorption of A is strong or P_A is high. (See section 2a)

c) How can you determine experimentally if the interaction between A and C still follows Langmuir behaviour at significantly much lower pressure values?

First it is necessary to record the data of P_A and t at much lower pressure.

At very low pressures, if there is a Langmuir behaviour, then $\ln P_A$ vs t should give a linear equation. (see section 2a)

3) The reaction $D(g) \rightarrow E(g)$ is catalysed by a solid, $F(s)$. The adsorption of D on F follows Langmuir behaviour.

a) How can you calculate the reaction rate constant, k , and the adsorption equilibrium constant for D on F, $K_{D/F}$, through experimental data? Explain how you would design the experiments, gather data and calculate the constants.

Data of P_D and time (t) needs to be collected for a range of Pressures (from high to low values of pressure) that represents Langmuir behaviour of the reactant.

To determine the rate constant k and the equilibrium constant K , you focus on the special cases (see question 2a)

When the adsorption of D is strong or P_D is high, then $P_D = P_D^0 - kt$, P_D vs t is a linear equation where the slope is k .

When adsorption of D is weak, or P_D is low, then $\ln P_D = \ln P_D^0 - kKt$, $\ln P_D$ vs t is a linear equation where kK is the slope, k is calculated from the previous equations at high P_D . Once you know k , then from the slope you can obtain K .

b) The reaction above is known to be inhibited by $X(g)$. X does not participate in the reaction, and the interaction with F is reversible. Explain how you would calculate the adsorption equilibrium constant for X on F , K_x from experimental data. Explain how you would design the experiments, gather data and calculate K_x .

$$\frac{KP}{1 + KP} = f$$

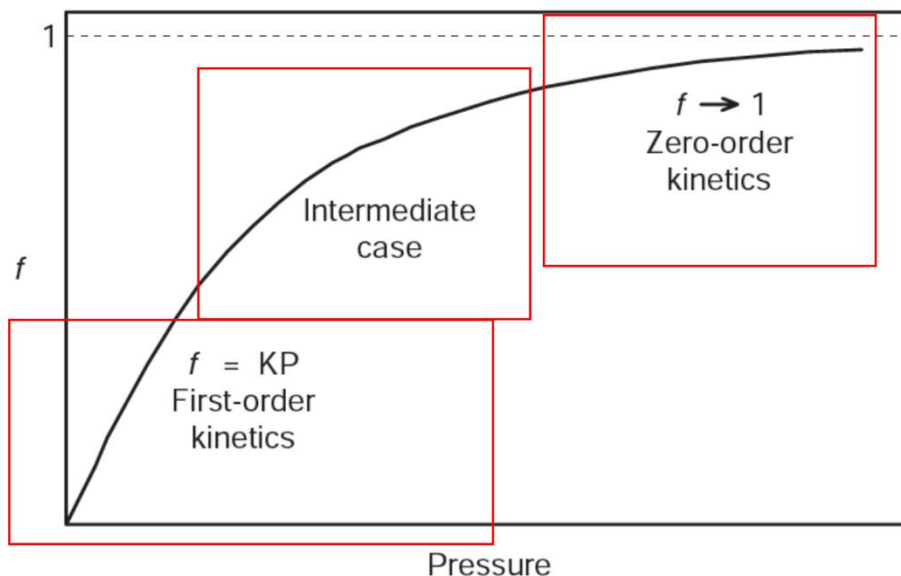
P_x and f are related through the Langmuir isotherm equation.

Experimentally you need to be able to collect information on the fraction of solid covered (f) by X and P_x .

From the Langmuir equation if you calculate the inverse of $\frac{1}{f} = \frac{1}{K} \times \frac{1}{P}$

The representation of $1/f$ vs $1/P$ gives a straight line with a slope being $1/K$

Alternatively. Assuming Langmuir behaviour at low P_x , $KP \ll 1$ then $f = KP$, if you plot f vs P , at low pressure the relationship between P and f is linear and the slope is K .



c) Explain why in b, it is important to know that X does not participate in the reaction?

It is important because if X does not participate in the reaction, we are talking about a process of physisorption, the process is reversible and we can use the Langmuir isotherm equation.

4) Propose how you would calculate the rate constant for a reaction between a solid G and a gas H, that forms a solid I on the surface of G: $G(s) + H(g) \rightarrow I(s)$. Explain how you would design the experiments, gather data and calculate the constants. Explain what assumptions are you making to answer this question.

- Reaction occurs at the interface, and the product forms a layer at the surface of the original solid.
- The reaction rate can be described mathematically based on the thickness of the product
- As x increases, it is more difficult for the gas to diffuse through the product to reach the reactant (the metal), therefore, the rate is proportional to $1/x$, therefore the rate law or rate equation becomes:

$$\frac{dx}{dt} = \frac{k}{x}$$

Rearranging:

$$x dx = k dt$$

Integrating:

$$\int x dx = \int k dt$$

Integrating



$$\frac{x^2}{2} = kt$$

Rearranging:

$$x = \sqrt{2kt}$$

Need to collect data of the thickness of the solid form on top (X) in time t. X vs t has a parabolic behaviour where you can elucidate the value of k.

